

The Kardar–Parisi–Zhang equation

From growing fronts to universality

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A colony boundary is irregular because local growth events fluctuate. A useful theoretical question is whether its large-scale roughness forgets the microscopic growth rules. The KPZ equation gives one possible answer. This note treats a one-dimensional interface in a two-dimensional setting, usually called $1 + 1$ dimensions.

1 Why a slope-squared term appears

Let $h(x, t)$ be the single-valued interface height. For local growth, reflection symmetry in x , and invariance under a uniform height shift, a leading long-wavelength equation is

$$\partial_t h = v_0 + \nu \partial_x^2 h + \frac{\lambda}{2} (\partial_x h)^2 + \eta(x, t), \quad \nu > 0. \quad (1)$$

The coefficient ν smooths curvature, λ measures slope-dependent growth, and v_0 is a constant drift. The zero-mean Gaussian noise is idealized as

$$\langle \eta(x, t) \eta(x', t') \rangle = 2D \delta(x - x') \delta(t - t'), \quad D > 0. \quad (2)$$

There is no equilibrium fluctuation–dissipation relation assumed between D and ν . If x and h both have length units, ν has units of length squared per time, λ of length per time, and D of length cubed per time.

For a front moving at constant normal speed v_n , its vertical speed is $v_n \sqrt{1 + (\partial_x h)^2}$. Expanding for small slopes produces $v_n + v_n (\partial_x h)^2 / 2$. This explains the allowed nonlinearity geometrically; a coarse-grained model need not have $\lambda = v_0$. The average speed also receives a contribution from slope fluctuations.

2 The observable is a scaling law

On a periodic interval of length L , remove the instantaneous mean height $\bar{h}(t) = L^{-1} \int_0^L h dx$ and define

$$W^2(L, t) = \left\langle \frac{1}{L} \int_0^L [h(x, t) - \bar{h}(t)]^2 dx \right\rangle. \quad (3)$$

Starting from a flat front, dynamic scaling takes the form, up to nonuniversal scale factors,

$$W(L, t) \sim L^\alpha f(t/L^z), \quad f(y) \sim \begin{cases} y^\beta, & y \ll 1, \\ \text{constant}, & y \gg 1, \end{cases} \quad \beta = \alpha/z. \quad (4)$$

Thus the growth regime has $W \sim t^\beta$, while the saturation time scales as L^z . An exponent extracted from a short time interval can instead describe a crossover or a finite-resolution artifact.

3 Two routes to the one-dimensional exponents

For the standard periodic KPZ equation, the stationary distribution of height differences, with the mean fixed, is

$$P_{\text{st}}[h] \propto \exp \left[-\frac{\nu}{2D} \int_0^L (\partial_x h)^2 dx \right]. \quad (5)$$

Its slopes are spatial white noise subject to the zero-mean-slope constraint, so height increments have Brownian scaling: $\alpha = 1/2$. In one dimension the KPZ nonlinearity leaves this stationary measure invariant. It nevertheless changes the dynamics.

After subtracting the constant drift, the equation also has statistical tilt symmetry. On an infinite interface, for constant slope u ,

$$h_u(x, t) = h(x + \lambda u t, t) + u x + \frac{\lambda u^2 t}{2} \quad (6)$$

obeys the same equation with translated noise. At the nonlinear KPZ fixed point this symmetry gives $\alpha + z = 2$. Combining the relations yields

$$\boxed{\alpha = \frac{1}{2}, \quad z = \frac{3}{2}, \quad \beta = \frac{1}{3}.} \quad (7)$$

By contrast, setting $\lambda = 0$ gives the Edwards–Wilkinson equation, with $\alpha = 1/2$ but $z = 2$ and $\beta = 1/4$ in one dimension. Equal-time roughness alone cannot distinguish the two dynamics. These exact KPZ exponents should not be carried over unchanged to a two-dimensional interface.

4 Connections to other theories, and to biology

Differentiating Eq. (1) and writing $u = \partial_x h$ gives a noisy viscous Burgers equation,

$$\partial_t u = \nu \partial_x^2 u + \lambda u \partial_x u + \partial_x \eta. \quad (8)$$

For smooth regularized noise and with v_0 removed, the Cole–Hopf variable $Z = \exp[\lambda h/(2\nu)]$ formally obeys

$$\partial_t Z = \nu \partial_x^2 Z + \frac{\lambda}{2\nu} \eta Z. \quad (9)$$

This connects interface growth to diffusion in a random environment and, through a path integral, to directed polymers. In the space–time white-noise limit, $(\partial_x h)^2$ is singular: the Cole–Hopf construction involves an Itô stochastic heat equation and a renormalized height drift. The smooth-noise chain rule is not a complete definition of that limit.

For a microbial front, local reproduction with short-range fluctuations can motivate a KPZ description. Nutrient-mediated nonlocal growth, quenched obstacles, mechanical coupling, or overhangs can invalidate it. A useful comparison checks spatial roughness, temporal growth, and finite-size scaling together; the geometry and initial condition also affect fluctuation distributions [2].

References

- [1] M. Kardar, G. Parisi, and Y.-C. Zhang, “Dynamic Scaling of Growing Interfaces,” *Phys. Rev. Lett.* **56**, 889 (1986). [doi:10.1103/PhysRevLett.56.889](https://doi.org/10.1103/PhysRevLett.56.889).
- [2] K. A. Takeuchi, *An appetizer to modern developments on the Kardar–Parisi–Zhang universality class*. [arXiv:1708.06060](https://arxiv.org/abs/1708.06060).